

MACHINE LEARNING PROJECT

Bank Marketing Campaign Deposit Subscription Prediction

Predicting whether a customer will subscribe to a term deposit using EDA and six supervised Machine Learning models

Team

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THE IDEA

Why This Project?

Banks run costly outbound campaigns to convince clients to open a term deposit. Calling every customer wastes time and money — most people say no.

Our goal: build a Machine Learning model that predicts, which customers are most likely to subscribe — so the bank can focus its effort where it counts.

PROJECT AT A GLANCE

11,162	customer records analyzed
17	raw features per customer
8	models built & compared
Recall	chosen as the priority metric

Features We Worked With

bank.csv — direct marketing data from a Portuguese bank, target column: deposit (yes/no)



Client Profile

- age
- job
- marital
- education
- default (credit)
- balance
- housing loan
- personal loan



Current Campaign

- contact type
- day
- month
- duration
- campaign (contacts made)



Previous Campaign

- pdays (days since last contact)
- previous (past contacts)
- poutcome (past result)

THE TEAM

Who Did What

F

Fatma

Exploratory Data Analysis (EDA) & Data Cleaning

R

Rehab

Logistic Regression & K-Nearest Neighbors (KNN)

H

Hagar

Decision Tree & Random Forest Models

M

Marwan

Boosting Models (XGBoost, LightGBM)

A

Abubakr

Ensamble method & Final Model Comparison

Key Findings from Exploration

1 Hidden Missing Values

job, education, contact and poutcome used the label "unknown" instead of real NaNs — treated as a category.

2 The pdays Trap

-1 meant "never contacted", not a real day count — replaced with a was_contacted_before flag.

3 Balanced Target

deposit is close to a 50/50 split between yes and no, so accuracy alone is a fair-but-not-sufficient metric.

4 Job Type Matters

students and retirees show the highest subscription rates; blue-collar and entrepreneur the lowest.

5 Past Success Predicts Future Success

customers with poutcome = success were far more likely to subscribe again.

Preprocessing & Feature Engineering

1

Fix pdays

Created was_contacted_before flag; replaced sentinel -1 with 0.

2

Cap Outliers

Capped campaign at the 99th percentile instead of deleting rows.

3

Encode

Binary yes/no columns mapped to 1/0; nominal columns one-hot encoded.

4

Split First

Train/test split done before encoding & scaling to avoid leakage.

5

Scale

StandardScaler fit only on training features, applied to test set.

Six Algorithms, One Goal

Every model was evaluated on Accuracy, Precision, Recall, F1-score and ROC-AUC, with Recall prioritized since missing a real subscriber is costlier than an extra call.

1 Logistic Regression

Linear baseline; tested default vs. class-balanced weighting.

2 K-Nearest Neighbors

Tuned K, distance metric and weighting via cross-validation.

3 Decision Tree

Tuned depth, split/leaf size and pruning (ccp_alpha) for Recall.

4 Random Forest

Ensemble of trees; tuned via RandomizedSearchCV for Recall.

5 XGBoost

Gradient boosting with scale_pos_weight for class imbalance.

6 LightGBM

Fast gradient boosting; final model in the RF + LightGBM ensemble.

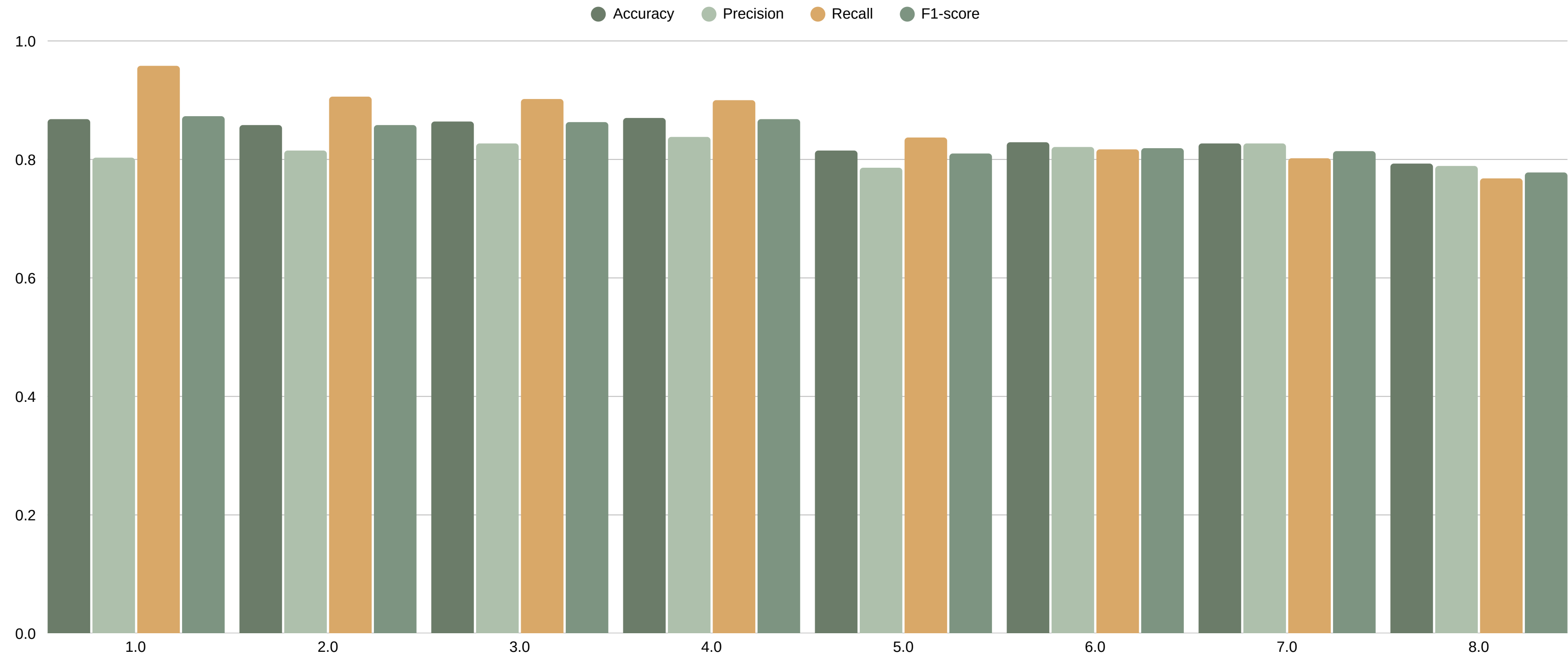
RESULTS

Final Model Comparison

Sorted by Recall — our primary metric

Model	Accuracy	Precision	Recall	F1	ROC-AUC
RF + LightGBM Ensemble	0.868	0.803	0.958	0.873	0.929
Random Forest (Recall-Opt.)	0.858	0.815	0.906	0.858	0.923
LightGBM (Recall-Opt.)	0.864	0.827	0.902	0.863	0.932
XGBoost (Recall-Opt.)	0.870	0.838	0.900	0.868	0.928
Decision Tree (Final)	0.815	0.786	0.837	0.810	0.892
Logistic Regression (Balanced)	0.829	0.821	0.817	0.819	0.907
Logistic Regression (Default)	0.827	0.827	0.802	0.814	0.907
KNN (Tuned)	0.793	0.789	0.768	0.778	0.874

Model Performance at a Glance



THE WINNER

RF + LightGBM Ensemble

Soft-voting ensemble of the two strongest tree-based models, threshold-tuned for accuracy.

95.8%

Recall

86.8%

Accuracy

87.3%

F1-score

Why Recall came first

- Missing a real subscriber costs a lost sale; an extra unnecessary call costs a few minutes.
- The ensemble catches 96 out of every 100 real subscribers in the test set.
- It still keeps accuracy and precision competitive with the single-model results.

RECALL RANKING

RF + LightGBM Ensemble



Random Forest



LightGBM



XGBoost



Decision Tree



Pipeline

Model Pipeline

- 1 Load & clean bank.csv
- 2 Engineer was_contacted_before, cap outliers
- 3 Train/test split (80/20, stratified)
- 4 One-hot encode + StandardScaler (fit on train only)
- 5 Train & tune 6 models with GridSearch / RandomizedSearchCV
- 6 Score on Accuracy, Precision, Recall, F1, ROC-AUC
- 7 Build RF + LightGBM soft-voting ensemble
- 8 Tune decision threshold on a validation split

Thank You

Bank Marketing Campaign

OUR TEAM

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